Q. Explain multilevel and multidimensional association rule mining in Detail.

Ans (I) Multilevel Association Rule.

(i) Multilevel association Rule are a Type of data mining ~~Techn~~ Technique used to Discover Relationship or patter b/w items in a dataset.

(ii) In multilevel association Rule, the items are organized into multiple levels or Hierarchies. Each level Represent Different level of abstraction or granularity.

Eg: Image a Store where products are grouped into categories, subcategories and individual items.

For eg:

- level 1: Electronics
- level 2: Mobile phones
- level 3: iphone, Samsung Galaxy, etc.

Multilevel Association Rule look for patters at Different level of Hierachy.

- At level 1: We might find: "people who By Electronics also buy Accessories".
- At level 2: We Might find: "People who Buy Mobile phone also buy mobile phone cases"
- At level 3: We Might find: "people who buy iphones often buy Apple Airpods."

This approach Help to analyze data at different level ~~at~~ of Detail, giving better insights for Decision making.

It also uses algorithm like Apriori or FP-Growth to identify which product are Often Brought Together.

~~Multi level Associa~~

MultiDimensional Association Rule

- Multidimensional Association Rule, also know as MultiDimensional association Rule or MDAR, are Type of Association Rule mining Technique that Consider ~~multidimen~~ multiple dimensional or attributes in data set.
- The goal to multidimen-sional association rule is to Discover ~~associa~~ association b/w item of ~~Differ~~ multiple Dimensions or attributes

Eg:- Now image a store collect data not just about what people buy but also who ~~by~~ buys it, when and where.

Customer: Age, Gender, income
product: Electronics, clothing, food
Time: weekday, weekend.

Multidimensional association Rule Find petters across these Dimensionals. For example

- Young adult (age 18-25) buy gaming laptop on weekend
- "Families with kid by cereal and milk in morning".

This approach looks for Relationship b/w attributes, not just ~~approach~~ product.

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PRC

Q Every Data structure in data warehouse contains time element. why?

Ans. The time element is included in every data structure in a data warehouse bcz it play a key role in making ~~thea~~ the data more useful and meaningful.

1. Analyzing over time

→ Time Helps us see how things changes over days, moths or years. This let Business spot Trends, patterns or problem and make Better decision.

Eg:- "Sales are Higher in December every year".

2. Looking at the Past

• Times lets us keep records of what Happend in the past. This Helps to Compare Different periods and learn from History.

Eg:- "Last year's profits were lower than this year's".

3. Tracking changes

Including time means we can Track when data was updated or changed. This is useful for checking old records or fixing mistakes.

Eg:- " This product price ~~increase~~ was updated ~~to~~ last month".

4. Combining data

- When data comes from many sources, time helps us ~~the~~ match it up correctly. This ensure all the information fits together neatly.

Eg:- "Customer order from one system match delivery time from another system"

5. Speeding up searches

- Time is used to Break Big datasets into smaller pieces, like monthly or weekly chunks. This makes it faster to find and analyze the Right data.

a use to in How well ... E = —
2/3 of data ⇒ Train set
1/3 d t ⇒ Test set

Q What are various method for estimating a classifier's accuracy?

Ans Various Techniques to evaluate the ~~classifier~~ ~~accur~~ accuracy of classifiers.

(A) Hold out

In Hold out method, the largest data set is Randomly divided into Three subsets:

1. Training ~~Test~~ set

- This the first part of the data. It is used to Teach the models.

2. Validation ~~Fo~~ set

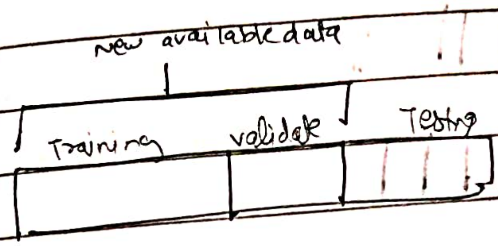
- This is the Second part of data. It's used to check how well ~~the~~ the model is learning and to fine-Tune it. Not every Model needs this step

3. Test set

- This ~~the~~ is the Third part of data. It use to see how well the model perform in front of new, unseen data. This give an idea of how the model will work in Real life.

Basicaly Two-Third of data are been allocated to Training set and Remaing One-Third is allocated to Test set.

$E = \frac{1}{K}\sum_{i=1} E_i$

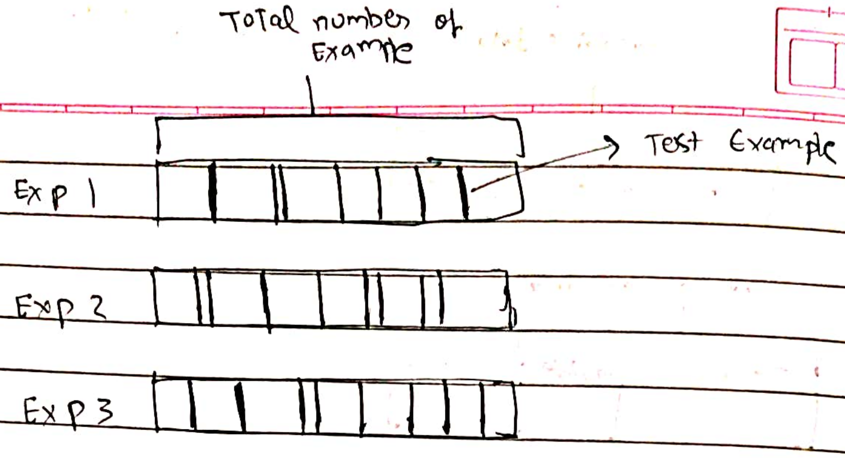


(B) Random Subsampling

- Random Subsampling is a variation of the Hold out Method. The Holdout method is been Repeated K-times
- The Holdout sub sampling involves randomly spliting the data into Training set and Test set
- On Training set, the data is been Trained and mean square ~~root~~ error (MSE) is been obtained from prediction on Test set.
- The Mean square Error (MSE) is Dependent upon split, this method is not Recommend. So a new split can give new MSE.
- The overall accuracy is been calculated as

$$\cancel{E = \frac{1}{k}\sum_{i=1}^{k} E(i)}$$

$$E = \frac{1}{K}\sum_{i=1}^{k} (E_i)$$



(c) Cross validation

- K-Fold Cross validation splits data into K groups. The model is Trained and Tested K times, ensuring every point is used once for Testing and Training.
- The data is Randomly Divided into K equal parts knows as folds.

Example : if K=5, then data is splits into 5 groups: D1, D2, D3, D4, D5.

- The model is Trained and Tested K times. Each time One fold is used as Test ~~~~ set (eg D1 is the First Round). The Remaining Fold are combined in Training set (eg D2, D3, D4 & D5 in First Round).
- These process is Repeated ~~for~~ so that each fold get test exactly one time.

~~(5) Boot strapping~~

Q Discuss Different Types of attributes?

Ans Types of attributes

This is the first step of Data Data preprocessing. we differentiate b/w Different type of attributes. ~~and~~

Description of attributes Types:

(1) Qualitative (Nominal (N), Ordinal (O), Binary (B)).

(2) Quantitative (Discrete, Continuous).

(I) Qualitative

(a) Nominal (N) → Related to Names.

- The values of a Nominal attributes are name of things, some kind of symbols.
- Nominal attributes Represents some category or state so it's also Referred as categorical attributes and there is no order (Rank, position) among value of nominal attributes.

Attributes	values
• colour	Black, Brown, white
• Categorical Data	lecture, proffessor, Assistant professor

(b) Binary Attributes

Binary attributes has 2 values/states. For example Yes or NO, affected or Unaffected, True or False.

(i) Symmetric :- Both values are equal important (Gender)
(ii) Unsymmetric :- Both values are Not equal important (Result)

Attributes	value
• Cancer Detected	Yes or No
• Result	Pass, fail
• Gender	Male ~~M~~, female

(c) Ordinal Attributes

• The ordinal Attributes contains values that have a meaningful sequence or Ranking b/w them.

Attribute	value
• Grade	A, B, C, D, E, F
• Basic pay scale	16, 17, 18

NA

DATE

(iii) Quantitative Attributes

- A ~~DE~~ numeric attribute is Quantitative bcz, it is a measurable Quantity, Represented in integer or Real values.

Two Main Types

(I) Discreate :- Discrete data have Finite value, it can be numerical and can also be in categorical form.

- The attribute is finite or countable infinite set of value.

Attribute	values
profession	Teacher, Buisness Man, peon,
Zip Code	301701, 110040, 400078,

(ii) Continuous :-

Continuous data have Infinit value, continuous data is of Float Type, There can be many values b/w 2 & 3.

Attribute	value
Height	5.4, 6.2, ---etc
weight	80, 33, ...etc

Q. Explain CLARANS extension in web mining?

Ans (i) CLARANS (Clustering Large Applications based on Randomized search) is an extension of clustering algorithm that is commoly used in web mining.

(ii) It is specially Designed to Handle large datasets and it is also Know for its efficiency and effectivness in Finding cluster.

(iii) CLARANS is a partial clustring algorithm aims to group similar data points together based on their attributes.

It works by Randomly selecting a data point as Initial Method for each cluster and iteratively swapping the medoids with other data points in the dataset.

(iv) The process ~~is~~ continues until a stopping criterion is met, such as maximum number of iteration or when algorithms converges.

Eg:- Imagine, you are organizing a messy Library, where books are scattered everywhere. You want to group similar Book together into section like Fiction, science, History, etc, but there are Tons of Books, and some are misplaced or don't fit perfectly into any category. This is were. CLARANS comes in its like a smart assistant that Helps to organize Library efficiently.

CLARANS in web mining offers several advantage:

1.) It can Handle large datasets efficiently, making it suitable for analyzing web-related data, which is often vast and complex.

2.) It is capable Of Handling noise and outliers, which are common in web data.

3.) Unlike some clustering methods, CLARANS can discover group of data in any shapes or pattern making it adaptable ~~to~~ and more flexible compared to other clustering algorithm.

So, CLARANS extension in web mining is a powerful tool for analyzing web-Related data.

Q. Explain Market Basket analysis with an Example.

Ans • Market Basket analysis is a Technique used to Find out which products are often Bought Together.
• It is based on concept that if a customer buy a certain item, They are Likely to buy another Related item as well.
• It Helps Business sell ~~~~ smarter and satisfy Customer Better!

Example :-

let's Consider a grocery store as an example. The store want to understand the purchasing Patter of its customer to improve marketing strategies.
By analyzing, the store can identify which item are Frequently Bought together.

Suppose store has Transcation data:

ID	Items
1	Bread, Milk
2	Bread, Egg
3	Milk, Egg
4	Bread, Milk, egg
5	

By seeing these, we can ~~item~~ identify following association!

1. Bread & Milk: These Two items are purchased Frequently together. So, if a Customer buys bread, there is a High chance they will also buy milk.

2. Bread ~~and~~ & Egg:- These Two items are also Frequenced Purcheed Together. So if a customer by Bread, there is a High chance they will ~~buy~~ buy Egg..

3. Egg & Milk:- These two items are also Frequently purchased together. So if a customer buy ~~the~~ Egg there is a High chance of Buying Milk also

Based on these association, the store can take various action to improve sales. For example they can place Bread, Milk ~~and~~ and Egg closer to each other in the store to encourage customer to buy all three items. They can also offer discounts or promotoins on these items to futher incentivize Customers.

By Using Market Basket analysis, business can Boost the sales by encourage customer to buy more items, Make shoopping easier and Customer are ~~satis~~ satisfied by getting Relevant items on there Place.

Q Illustrate major steps in ETL process.

Ans ETL Stands for Extract, Transform and Load and is a process for collating and unifying Different Data sets.

There are ToTal Fives steps in ETL process ie Extract, clean, Transform, Load and analyze.

(1.) Extract :

(i) Data set are captured from unstructured source and placed into Tempory staging location.

(ii) Extract Data from these sources. This can be done using various method such as SQL Quries from Database, API calls from web service or File Reading ~~from~~ for Flat files (CSV, JSON, etc).

(2) Clean :

i Before Data set can be Transformed, They must be cleaned and preprocessed Through series of Standardization and Normalization steps, which removes any Defects and invalid Records

E & D load to ...
... → Databa...
API calls → web serv
File read → Flatfi...

(3.) Transform:- The Data set are processed by applying a series of Rules, Then converted into correct Format for Intended destination.

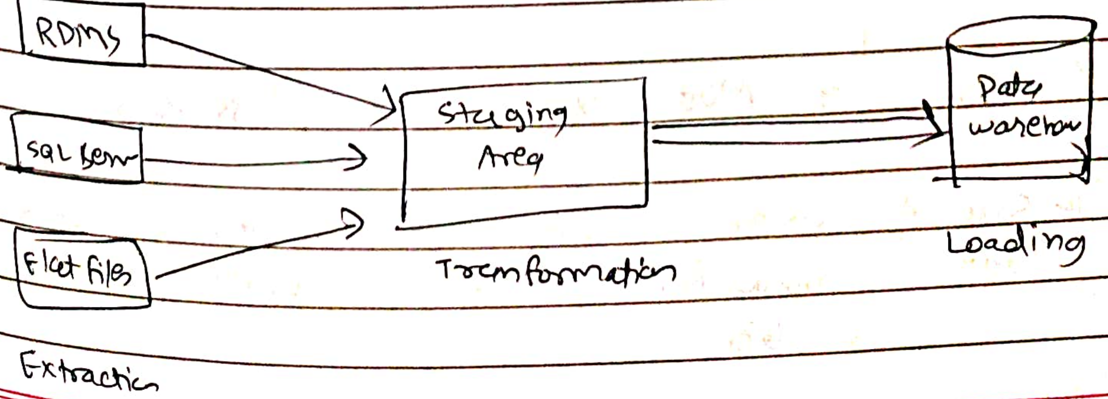
- The Three most common Transformation in ETL process are Key Restructuring, Data Filtering and Derivation.

(4) Load :- Load the Transformed data into Target System. This could be data warehouse, database or another storage system.

- Ensure that the Data loaded into Target System is complete and accurate.
- Index the Data if necessary to ~~impo~~ improve Query performance.

(5) Analyze :- In last stage, the data is now Strategic asset. It is visualized using leading BI Tools To Deliver Reports to the Business ~~B~~ Team.

The ETL process is an iterative process that is Repeated as new data is added to the warehouse. The process is important bcz it ensure that data in data warehouse is accurate, complete and up-to-date.



Q Differentiate ER Modelling & Dimensional Modelling?

Ans

Parameters	ER modelling	Dimensional modelling
1. Orientation	1. Transcition oriented	1. Subject oriented
2. Component	2. Entities & Relation ship	2. Fact Table & Dimension Table.
3. Granularity	3. Few level of Granularity	3. Multiple level of Granularity.
4. Type of Information	4. Real Time Information	4. Historical Information
5. Volatility	5. Highly volatily Data	5. Non-volatile data
6. Model Type	6. Physical & Logical Model	6 physical Model
7. Normalization	7. Normalization is suggested	7. De-Normalization is suggested
8. Application Type	8. OLTP Application	8. OLAP Applicatin
9. Users	9. More users	9. less users
10. Size of data	10. size of data varies from MBs to GBs	10. size of data varies from GBs to TBs